Number Theory

Factorization

n

all factors

what are the factors of a no. 'n' ?

$$f \rightarrow n \equiv (n \% f == 0)$$

n iterations

'n'

(1 . . . . . n) → for(i=1; i<=n; i++)

if(n % i == 0)

print(i);

i = 1 . . . n

(n % i == 0) → print all those 'i'.

O(n) → time complexity

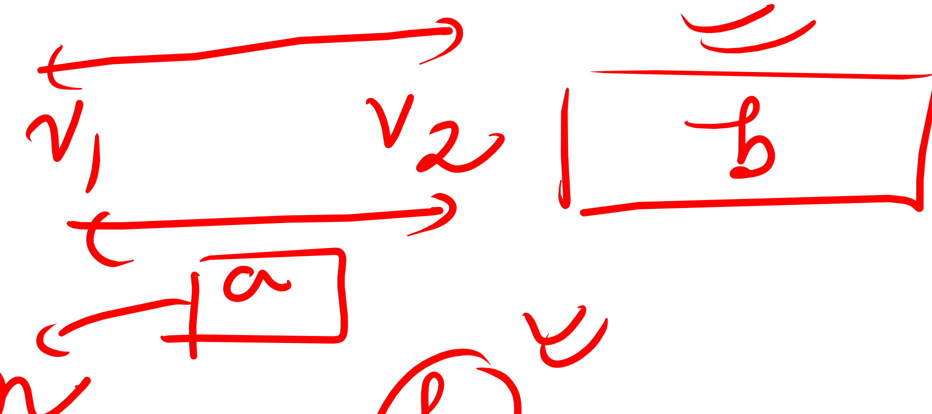
div = a | b

b | a

a → 'n'
↳ is a factor

$n/a = b \Rightarrow \frac{n}{b} = a$

$n = a \cdot b$



$n/a = b$

$n = 8$

1, 2, 4, 8

$8 = 1 * 8$

$8 = 2 * 4$

$n/s = s'$

$s <= s'$

b → smallest

a greatest

a

b

$n = a \cdot b$?

$25 = 5 * 5$

a → max limit

b

$n = a^2$

$a \to \sqrt{n}$

$a \to 1 \ldots \sqrt{n}$

$a \to 1 \ldots \sqrt{n}$

~~a →~~

$n = a^2$

$a^2 = n$

$a = \sqrt{n}$

i

$a \to 1 \ldots \sqrt{n}$

$\sqrt{n+1} \ldots n$

i → factor

$n \% i = 0$

n/i → factor

$n/\sqrt{i} = i$

$n/i \to \sqrt{i}$

$i <= \sqrt{n}$

i

$1 \ldots \sqrt{n} \Rightarrow O(\sqrt{n})$

$\boxed{\text{sqrt}(n)}$ $\otimes$

$\sqrt{n} < n$

$O(\sqrt{n})$

$i \rightarrow$ factor

$i <= \sqrt{n}$

$i^2 <= n$

i

$1 \ldots n$

$(i <= \sqrt{n})$

$i^2 <= n$

$i \rightarrow 1 \ldots \sqrt{n}$

$i * i <= n;$

i ✓

$\cancel{n \% i}$

$n = 25$

5 5

b → $\cancel{n / i}$

b → $25/5 = 5$

```
for(i=1; i*i<=n; i++){
    if(n%i==0){
        print(i);
        if(n/i != i)
            print(n/i);
    }
}
```

$O(n) \longrightarrow O(\sqrt{n})$

$O(N\sqrt{N})$

$N \to 10^4$

$\sqrt{10^4} = 10^2$

$O(N\sqrt{N})$

$N\sqrt{N} = 10^6 <= 10^8$

All primes from $(1 \leftrightarrow N) \to i$

$1 \ldots N$

$i \to$ factor check

2 factors

prime

$i \to 1 \ldots N$

$\sqrt{N}$

$O(N\sqrt{N})$

$1 \ldots N$

count of factor

$\sqrt{N}$

$O(N * \sqrt{N})$

prime → 1, num
2 3 4 ...

Sieve of eratosthenes

N = 10

primes[N+1];

0	1	2	3	4	5	6	7	8	9	10
~~T~~	~~T~~	T	T	~~T~~	T	~~T~~	T	~~T~~	~~T~~	~~T~~
F	F			F		F		F	F	F

p → T

p2

prime = 2

multiples of 2 → not a prime

p → 1*

25

49

5

25

10 F

15 F

20 F

p^2

p^2 →

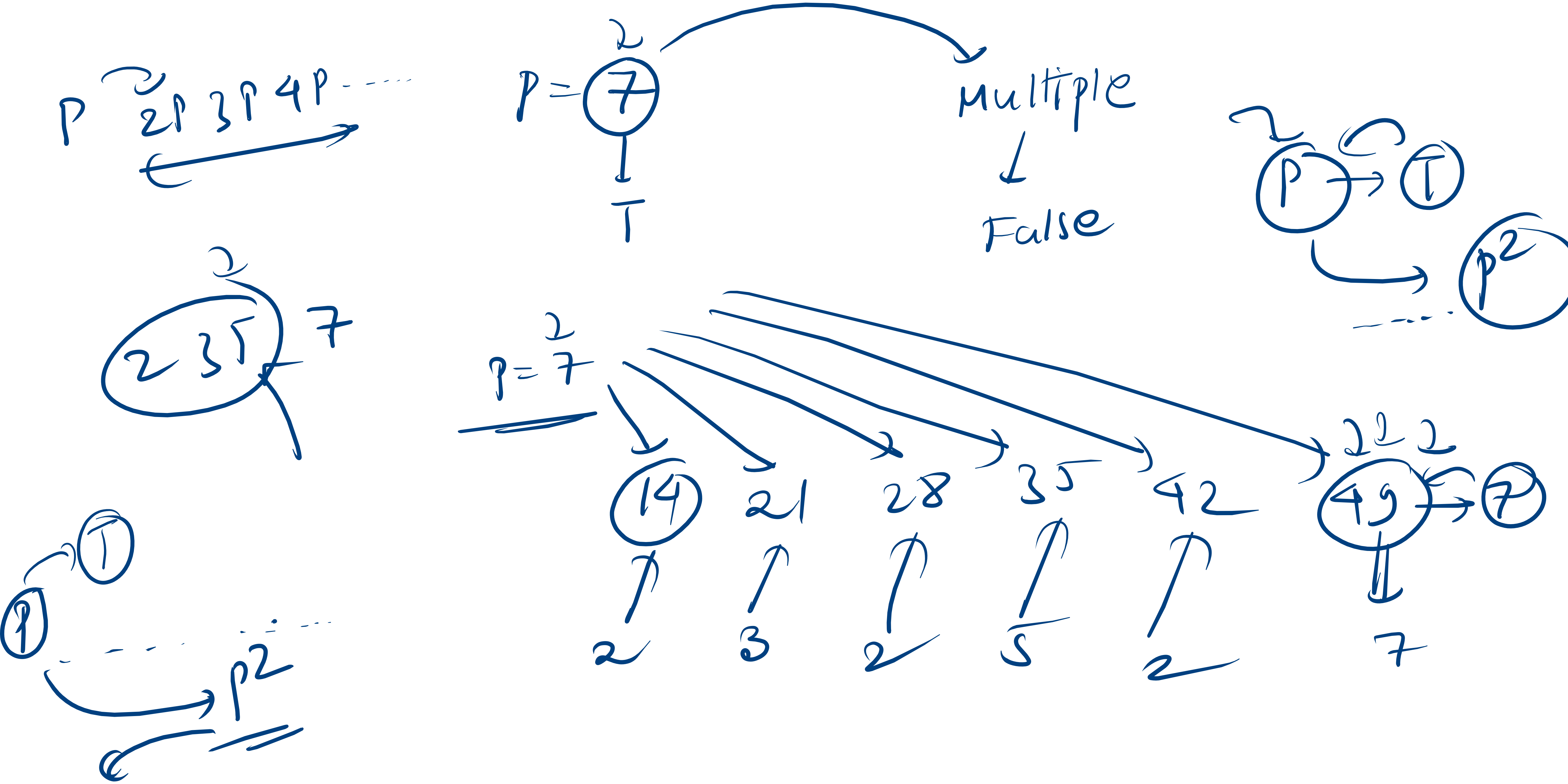
7

14 21 28 35 42 49

2 3 2 5 2

F

1/10


P 2P 3P 4P
p = 7
T
Multiple
False
P
T
p²
2 3 5 7
p = 7
14 21 28 35 42 49
2 3 2 5 2 7
7
P
T
p²

p → p^2 ...... multiple → false.

$\sqrt{N}$

$$\begin{cases} p \to \sqrt{N} \\ p^2 \to N \end{cases}$$

$1 \; . \; . \; . \; N$

$p \to \{1 \; . \; . \; . \; \sqrt{N}\}$

$p \to p^2$

$p^2 \leq N$